

2022-2 Calculus II (06)

Midterm Exam

2022.10.25, 15:00–16:30

9 problems. All your solutions should contain rigorous explanation.

1. Let $f(x, y) = \ln(x - 2y)$.

- (a) (3pts) Find the domain and range of f .
- (b) (3pts) Sketch any three level curves of f .
- (c) (4pts) Let (a, b) be a fixed point in the domain of f . Find the maximum value of the directional derivative $D_{\mathbf{u}}f(a, b)$ and the unit vector $\mathbf{u}$ maximizing $D_{\mathbf{u}}f(a, b)$.
- (d) (6pts) Find equations of the tangent plane and normal line to the surface $z = f(x, y)$ at $(3, 1, 0)$.

2. Let

$$f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

- (1) (4pts) Find $f_x(0, 0)$ and $f_y(0, 0)$, if they exist.
 - (2) (4pts) Let $\mathbf{u} = \langle a, b \rangle$ be a unit vector with $a, b \neq 0$. Find $D_{\mathbf{u}}f(0, 0)$, if it exists.
 - (3) (6pts) Is f_x continuous at $(0, 0)$? Justify your answer.
3. (10pts) Suppose the volume of a rectangular box is 4. Does the minimum of surface area of such a box **without a lid** exist? If it exists, find the minimum.
4. (10pts) Find the maximum and minimum values of f subject to both constraints.

$$f(x, y, z) = x^2 + y^2 + z^2; \quad x - y = \frac{1}{2}, \quad y^2 + z^2 = 1$$

5. (10pts) Evaluate the double integral

$$\iint_D e^{y^{3/2}} dA,$$

where D is the region in $\mathbb{R}^2$ bounded by the y -axis, the line $y = 1$ and the curve $y = x^2$.

6. (10pts) Find the volume of the solid enclosed by the surface $z = -x^2 + xy^2$ and the planes $z = 0$ and $y = 1$.

7. (10pts) Evaluate the integral

$$\int_0^{\frac{\sqrt{2}}{2}} \int_{\arcsin y}^{\frac{\pi}{4}} \cos x \sqrt{1 - \cos^2 x} \, dx dy$$

8. (10pts) Find the volume of the solid that lies under the paraboloid $z = \sqrt{x^2 + y^2}$, above the xy -plane, and inside the cylinder $x^2 + y^2 = 4y$.
9. (10pts) Find area of the part of the cylinder $x^2 + z^2 = 1$ that lies above the xy -plane and inside the cylinder $x^2 + y^2 = 1$. (Hint: $\int \frac{1}{1+\sin \theta} d\theta = -\frac{2}{1+\tan(\frac{\theta}{2})} + C$)